

# Infosys Internship - Individual Contribution Documentation

- Penchala Prasad

## 1. Role and Responsibility

As a member of the project development team, my primary responsibilities were related to frontend development and UI integration. I worked on establishing the user interface, configuring the application routing, and ensuring the application provided a responsive experience. My contribution involved building the frontend application using React and Tailwind CSS for presenting and interacting with project data.

## Frontend Development and Integration

My major contributions to the project were building the application interface with **React (JSX)** and **Tailwind CSS**. The frontend was required to present data to the user, handle interactions, and provide role-based access portals. I worked on establishing the visual layout and verified that the application components could render and communicate state successfully.

The major tasks performed were:

- Set up the React environment using Vite for the project.
- Created the required frontend components, pages, and assets.
- Configured the application routing using React Router v7 with protected route wrapping.
- Styled the user interface dynamically using Tailwind CSS v4.
- Created interactive data visualizations and charts using Recharts.
- Tested user interface operations and responsive design.
- Verified that application state could be managed and retrieved successfully via LocalStorage.
- Checked the rendered UI components using browser developer tools.

## React & Tailwind CSS

I used **React (JSX)** as the primary library for building the user interface and **Tailwind CSS** for rapid, utility-first styling. React and Tailwind CSS helped me to visually construct the dashboard structure and the data exhibited by the application.

The activities performed using these tools included:

- Building the themed login portal and dynamic dual-tab settings.
- Creating interactive components like patient registries and analytical graphs.
- Viewing the available UI components and layouts.
- Inspecting DOM elements rendered on the screen.
- Verifying whether user input submitted through the application updated the state correctly.

- Checking the structure and properties of the JSX components.
- Using developer tools to identify and troubleshoot styling or state-related issues.

This provided a convenient way to verify the user experience and interface of the application.

## Frontend Interaction Workflow

The frontend integration followed the general workflow below:

User → Frontend (React JSX) → State Update (LocalStorage) / API  
Request → Backend → API Response → Frontend State Update → UI  
Re-render

The user interacts with the frontend application (e.g., clicking a patient record) and submits the required action. The frontend updates the local state or sends the request to the backend. After processing the interaction, the frontend components dynamically re-render based on the new data or state, where the result (such as a patient worksheet or dynamic graph) is displayed to the user.

## Challenges Faced

During the development and integration process, I faced some challenges related to frontend architecture and user interface connectivity.

### Frontend Challenges:

- Configuring the React Router v7 protected routes correctly.
- Understanding the component lifecycle and state synchronization using LocalStorage.
- Verifying whether dynamic data visualizations (Recharts) rendered accurately across different screen sizes.
- Troubleshooting Tailwind CSS utility class conflicts and dynamic re-theming logic.
- Understanding the communication between the interactive UI components and the data layer.

These challenges helped me improve my understanding of full-stack application development and frontend integration.

## Overall Contribution

This project gave me practical exposure to the development lifecycle, collaborative software development, user interface implementation, frontend routing, testing, and debugging.